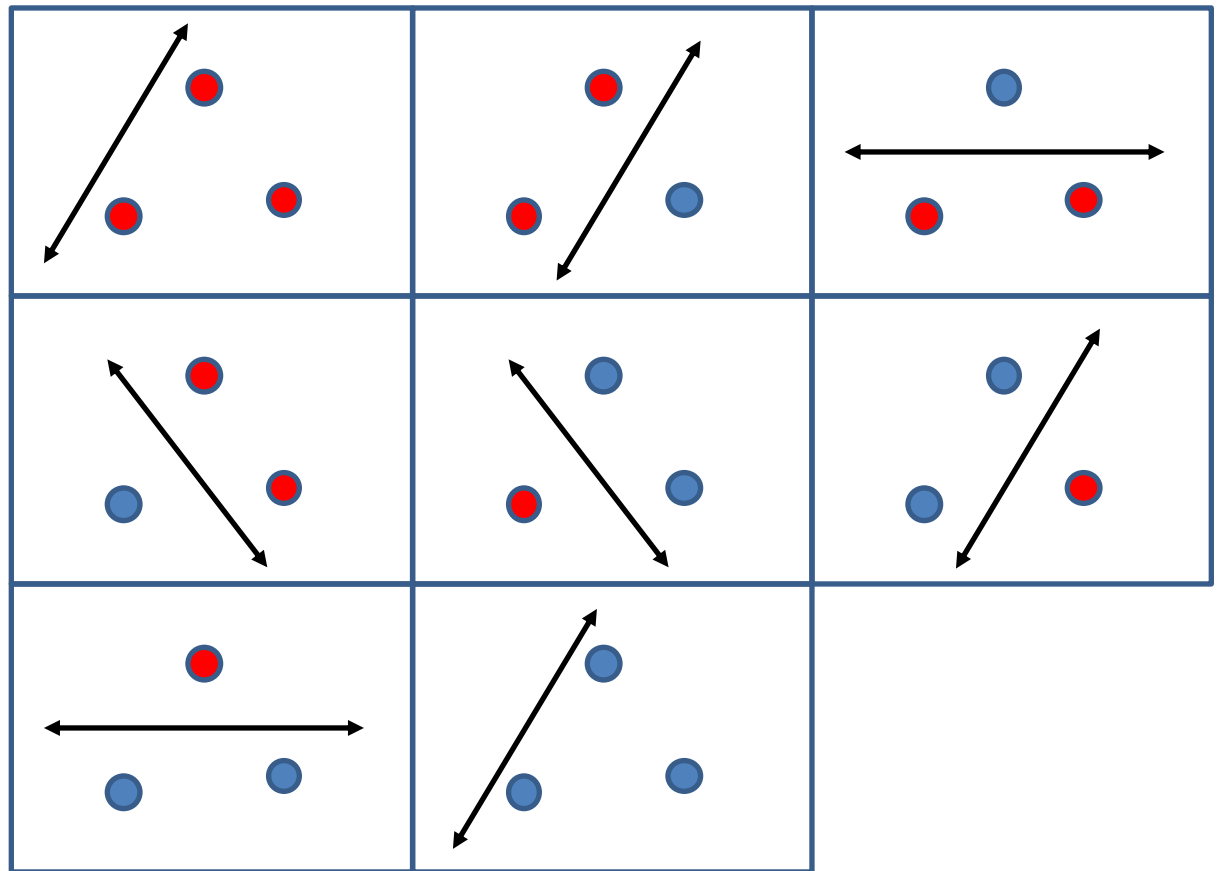
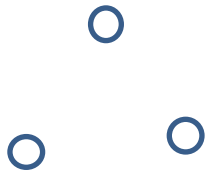


More on VC-dimension.

Lee-Ad Gottlieb

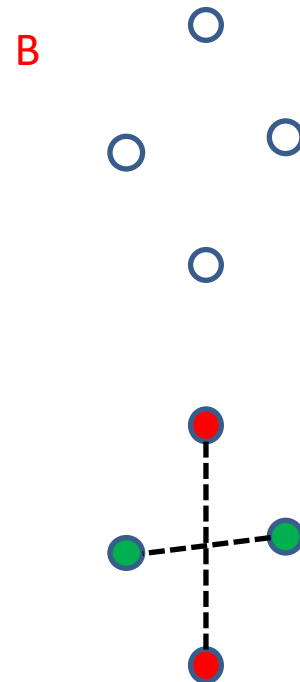
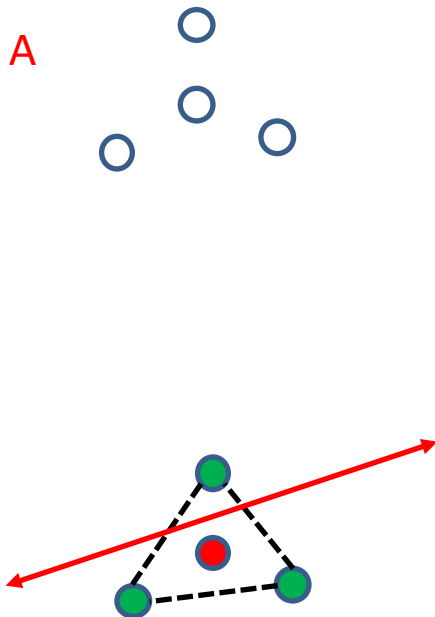
VC-dimension

- The infinite set of (**bi-directional**) lines can shatter **some** set of 3 points in 2D



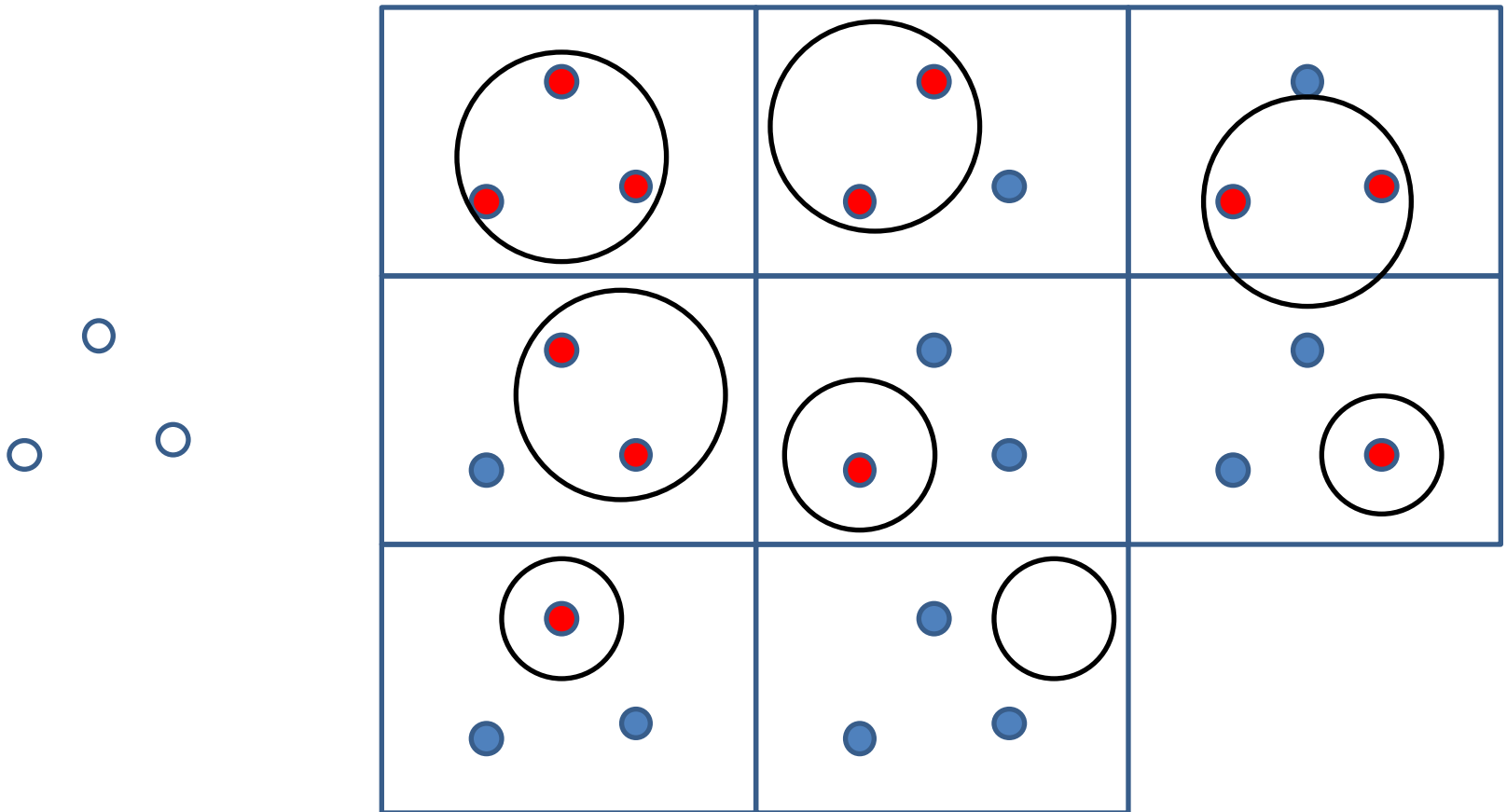
VC-dimension

- There does not exist *any* set of four points which can be shattered by the infinite set of lines.

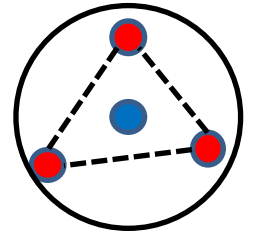


Balls in 2D

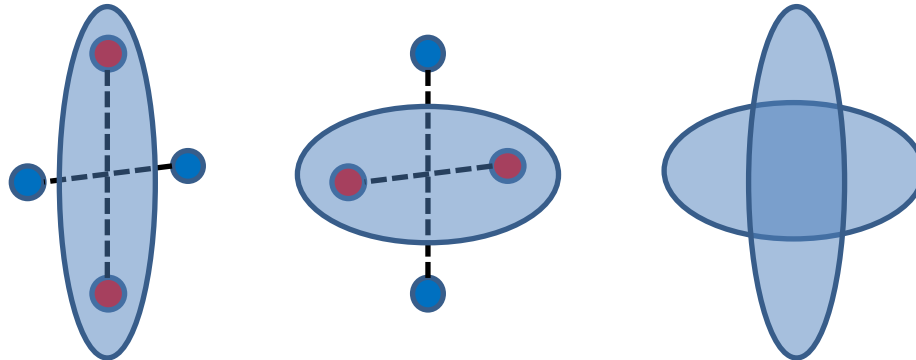
- Claim: VC-dimension of infinite set of uni-directional balls (inside=red) in 2D is 3. Lower-bound:



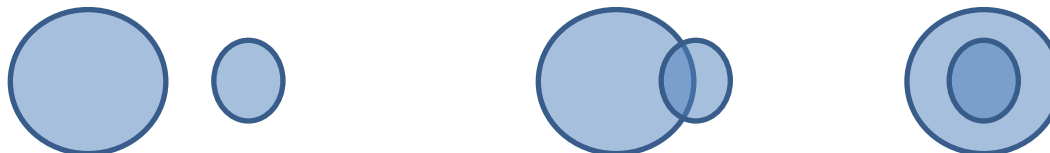
Balls in 2D



- Proof: Upper-bound
 - **A.** If one point is in the convex hull of the others, must have same label.
 - **B.** If points are in general position, find opposite pairs. There must be a circle containing each pair

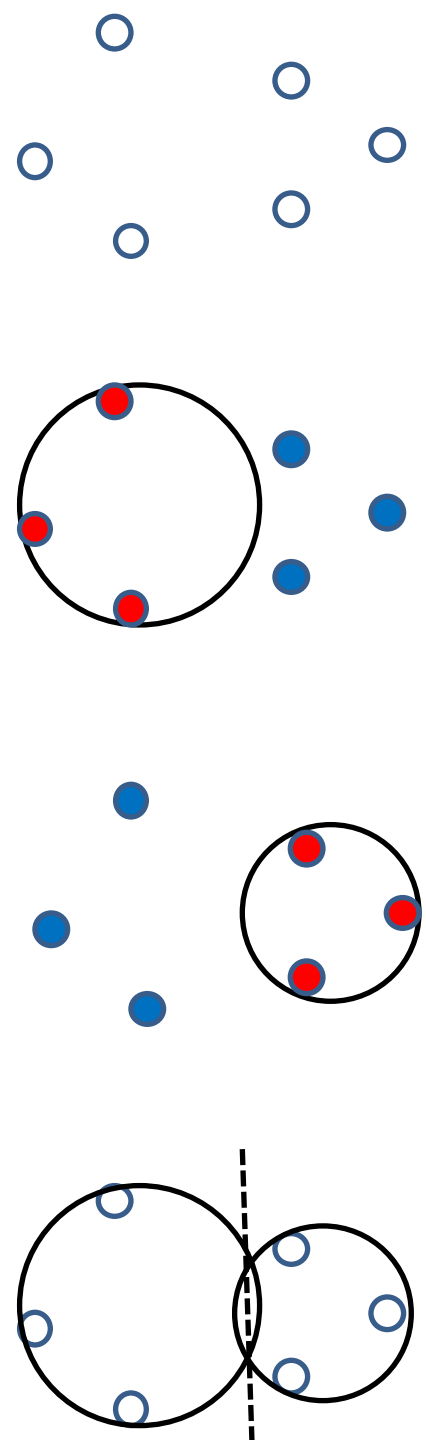


- But the symmetric difference of two circles is at most 2



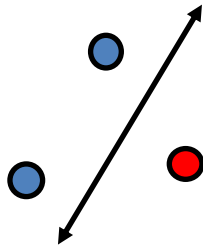
Balls in 2D

- Another proof for the upper-bound
 - Take a set shattered by circles.
 - For any division of the set into subsets A, B , there must be two circles covering A, B
 - The intersection of the two circles is a line, so the points are separated by a hyperplane
 - So circles can't shatter more points than hyperplanes

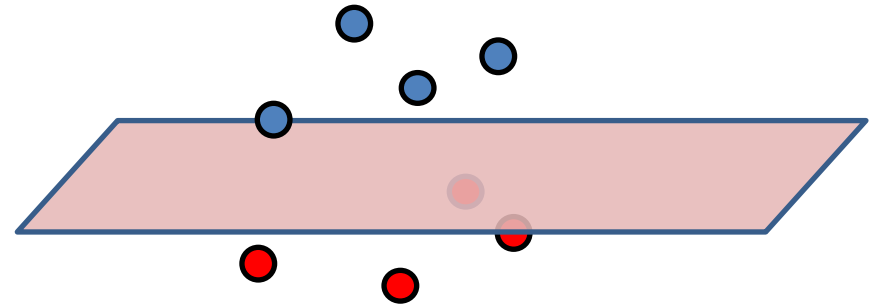


VC-dimension

- Claim: The set of $d-1$ dimensional hyperplanes has VC-dimension $d+1$ on $\mathbb{R}^d$.



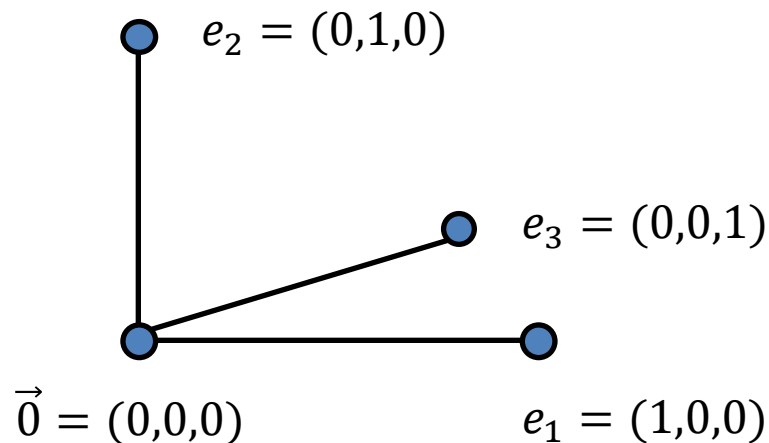
2D points



3D points

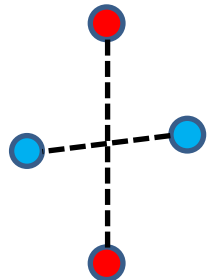
VC-dimension

- Lower-bound:
 - The set of $d-1$ dimensional hyperplanes can shatter $d+1$ points in $\mathbb{R}^d$.
 - Proof: $\vec{0}, e_1, e_2, \dots, e_d$



VC-dimension

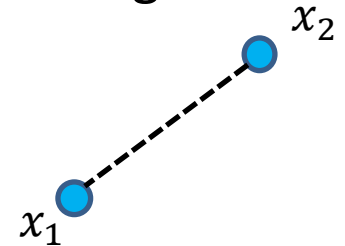
- Upper-bound:
 - The set of $d-1$ dimensional hyperplanes cannot shatter $d+2$ points in $\mathbb{R}^d$.
 - Proof:
 - Radon's theorem: any set of $d+2$ points in $\mathbb{R}^d$ can be partitioned into two disjoint sets whose convex hulls intersect.
 - The intersection point can't be both blue and red.



Background: convex hull

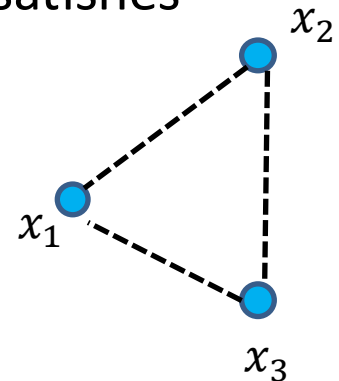
- Given two points x_1, x_2 every point on the segment connecting them satisfies

- $\alpha_1 x_1 + \alpha_2 x_2$
- where $\alpha_1 + \alpha_2 = 1$ and $\alpha_1, \alpha_2 \geq 0$



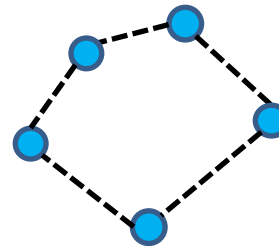
- Given three points x_1, x_2, x_3 every point in their triangle satisfies

- $\alpha_1 x_1 + \alpha_2 x_2 + \alpha_3 x_3$
- where $\alpha_1 + \alpha_2 + \alpha_3 = 1$ and $\alpha_1, \alpha_2, \alpha_3 \geq 0$



- Given n points $x_1, \dots, x_n$ every point in their hull satisfies

- $\sum_i \alpha_i x_i$
- where $\sum_i \alpha_i = 1$ and $\alpha_i \geq 0$



Radon's theorem

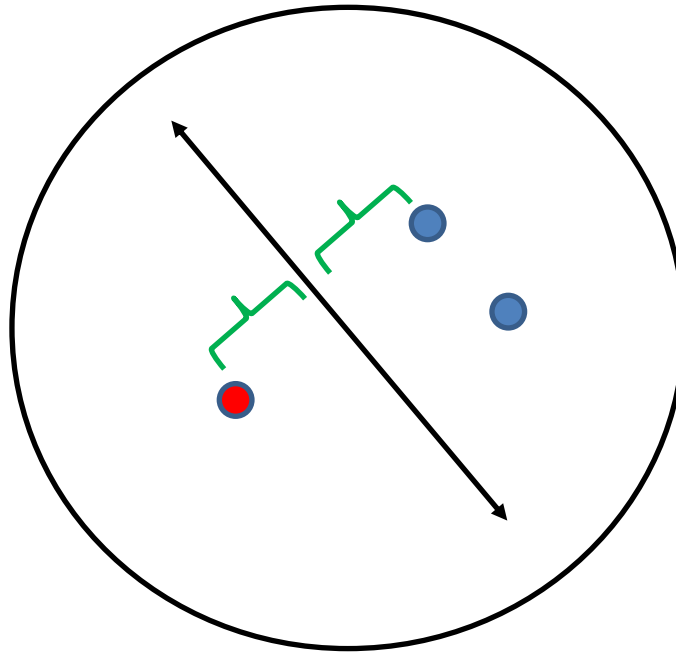
- Proof:
 - Let the points be $x_1, \dots, x_{d+2}$
 - Find non-zero constants $a_1, \dots, a_{d+2}$ satisfying
 - $\sum_{i=1}^{d+2} a_i x_i = \vec{0}$ $\leftarrow d$ equalities (dimensions)
 - $\sum_{i=1}^{d+2} a_i = 0$ $\leftarrow 1$ equality
 - $d+1$ equations with $d+2$ unknowns must have a solution.
 - If $a_i > 0$ put x_i into **I**, else put x_i into **J**
 - Convex hulls of **I, J** must intersect:
 - Set $A = \sum_{x_i \in I} a_i = \sum_{x_j \in J} -a_j$
 - $\sum_{x_i \in I} \frac{a_i}{A} = \sum_{x_j \in J} \frac{-a_j}{A} = 1$
 - $\sum_{x_i \in I} a_i x_i = \sum_{x_j \in J} -a_j x_j$
 - $\sum_{x_i \in I} \frac{a_i x_i}{A} = \sum_{x_j \in J} \frac{-a_j x_j}{A}$
 - This is a point inside both hulls

Follows from
 $\sum_{i=1}^{d+2} a_i = 0$

Follows from
 $\sum_{i=1}^{d+2} a_i x_i = \vec{0}$

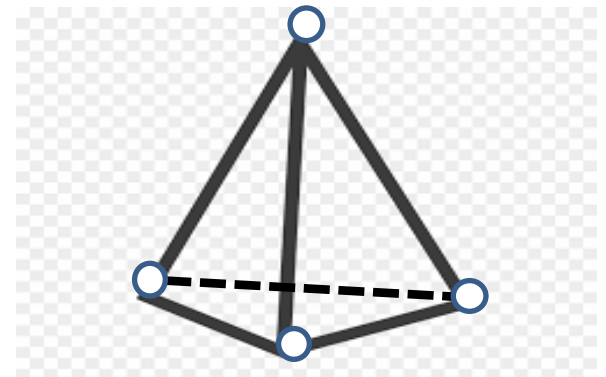
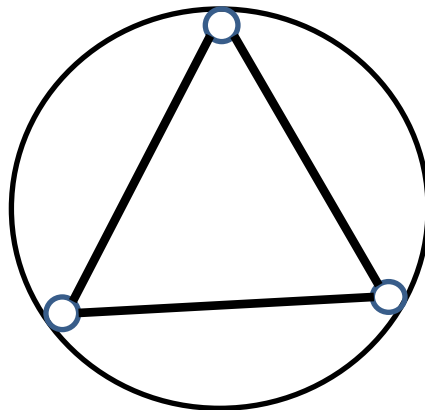
Planes with margin

- A hyperplane with **margin** γ can shatter only $O(R/\gamma)^2$ points within radius R , irrespective of the dimension.



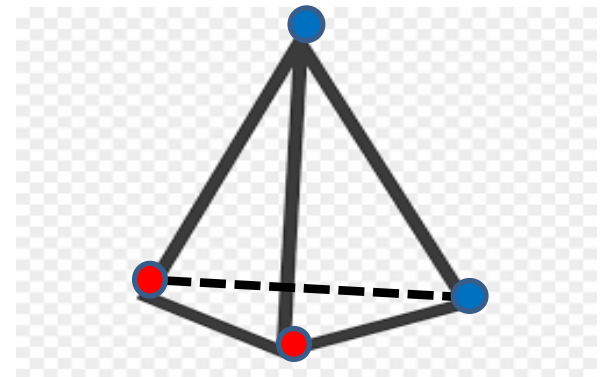
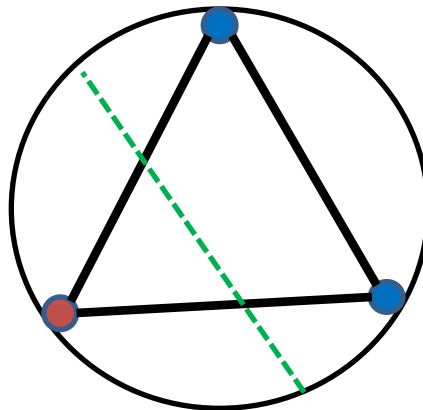
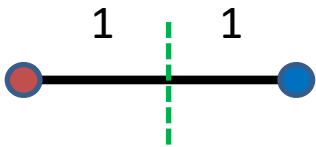
Planes with margin

- Intuition:
 - Largest margin achieved when points are at equal distance
 - For n points, use $n-1$ dimensions



Planes with margin

- Intuition:
 - Largest margin achieved when points are at equal distance
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Planes with margin

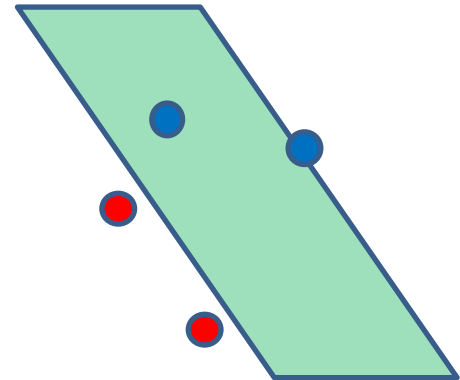
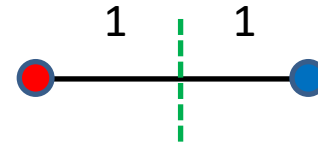
- Intuition, $R=1$

- 2 points: margin 1

- 4 points: margin $\frac{1}{\sqrt{2}}$

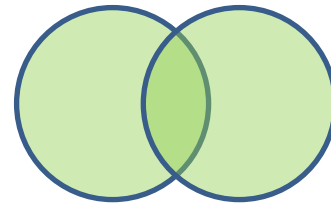
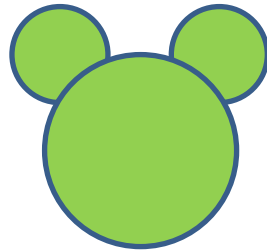
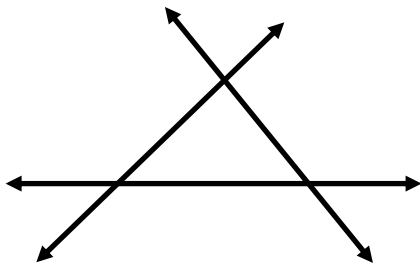
- k points: margin $\frac{1}{\sqrt{k/2}}$

- $\gamma \leq \frac{1}{\sqrt{k/2}} \quad \rightarrow \quad k \leq \frac{2}{\gamma^2}$



Union of hypotheses

- What about objects that are the union or intersection of simple rules?



Union of hypotheses

- Theorem:
 - Let $h_1, h_2, \dots \subseteq H$ be a class of hypotheses (rules) with VC-dimension d .
 - Let $H' = \{\bigcup_{i=1}^s h_{j_i}\}$
 - Then $\text{VC-dimension}(H') \leq 2ds \log_2(3s)$
- Note: Can prove same theorem with intersection ($\cap$) instead of union ($\cup$).

Union of hypotheses

- Proof:

- For any point-set P of size n

$$\Pi(H', P) \leq [\Pi(H, P)]^s \leq (en/d)^{ds}$$



Number of
possible labellings
in H , raised to the s



Sauer's
lemma

Union of hypotheses

- Proof:

- For any point-set P of size n

$$\Pi(H', P) \leq [\Pi(H, P)]^s < (en/d)^{ds}$$

- Let P be the largest point-set shattered by H' , so

$$2^n < (en/d)^{ds}$$

- Take $n \geq 2ds \log_2(3s)$:

$$2^{2ds \log_2(3s)} < (e2ds \log_2(3s)/d)^{ds} \rightarrow$$

$$\frac{9s}{2e} < \log_2(3s)$$

- Doesn't hold for $s \geq 2$

